CPP102 — WEEK 17

Simulated Device Monitoring and Alert System

Integrative Programming Project

Student Name	Cubil, Christian T.
Student Number	2403078
Section	BS 1CPE-B

Contents in order: 1) Python Program, 2) README.md, 3) test_cases.md

1. Python Program — device_monitor.py

Source code (save the project file as **device_monitor.py**).

```
# Simulated Device Monitoring and Alert System

def classify_reading(reading):
    """Classify one temperature reading."""
    if reading &lt; 30:
        return "NORMAL"
    elif reading &lt;= 35:
        return "WARNING"
    else:
        return "CRITICAL"

def view_readings(readings):
    """Show all temperature readings."""
    print("\nCURRENT READINGS")
    for i in range(len(readings)):
        print("Reading", i + 1, ":", readings[i], "°C")

def add_reading(readings):
    """Ask for a reading and add it if valid."""
    value = input("Enter new reading: ")

    if value == "":
        print("Invalid input. The reading cannot be empty.")
    elif value.startswith("-") and value[1:].isdigit():
        value = int(value)
        print("Invalid reading. Enter a value from 0 to 100.")
    elif not value.isdigit():
        print("Invalid input. Enter a whole number only.")
    else:
        value = int(value)

        if value &lt; 0 or value &gt; 100:
            print("Invalid reading. Enter a value from 0 to 100.")
        else:
            readings.append(value)
            print("Reading added successfully.")

def classify_readings(readings):
    """Show the category of every reading."""
    print("\nREADING CLASSIFICATION")
    for reading in readings:
        print(reading, "°C -&gt;", classify_reading(reading))

def generate_summary(readings):
    """Show the temperature summary and overall status."""
    total = 0
    smallest = readings[0]
    largest = readings[0]
    normal = 0
    warning = 0
    critical = 0

    for reading in readings:
        total = total + reading

        if reading &lt; smallest:
            smallest = reading
        if reading &gt; largest:
            largest = reading
```

```
        category = classify_reading(reading)
        if category == "NORMAL":
            normal = normal + 1
        elif category == "WARNING":
            warning = warning + 1
        else:
            critical = critical + 1

    count = len(readings)
    average = total / count

    print("\nMONITORING SUMMARY")
    print("Number of readings:", count)
    print("Total of readings:", total)
    print("Average temperature: {:.2f} °C".format(average))
    print("Minimum temperature:", smallest, "°C")
    print("Maximum temperature:", largest, "°C")
    print()
    print("NORMAL:", normal, "({:.2f}%)".format(normal / count * 100))
    print("WARNING:", warning, "({:.2f}%)".format(warning / count * 100))
    print("CRITICAL:", critical, "({:.2f}%)".format(critical / count * 100))

    if critical &gt; 0:
        print("Overall Status: ATTENTION REQUIRED")
    elif warning &gt; 0:
        print("Overall Status: MONITOR")
    else:
        print("Overall Status: NORMAL")

def main():
    """Display the menu until the user chooses Exit."""
    readings = [28, 31, 35, 37, 29]
    choice = ""

    while choice != "5":
        print("\nSIMULATED DEVICE MONITORING SYSTEM")
        print("----------------------------------")
        print("[1] View Readings")
        print("[2] Add Reading")
        print("[3] Classify Readings")
        print("[4] Generate Summary")
        print("[5] Exit")

        choice = input("Enter choice: ")

        if choice == "1":
            view_readings(readings)
        elif choice == "2":
            add_reading(readings)
        elif choice == "3":
            classify_readings(readings)
        elif choice == "4":
            generate_summary(readings)
        elif choice == "5":
            print("Thank you for using the Device Monitoring System.")
        else:
            print("Invalid choice. Please enter a number from 1 to 5.")

main()
```

2. README.md

Simulated Device Monitoring and Alert System

1. Project Title

Simulated Device Monitoring and Alert System

2. Student Name

Cubil, Christian T.

3. Student Number

2403078

4. Section

BS 1CPE-B

5. Project Description

This is a simple Python program that stores temperature readings and classifies each reading as NORMAL, WARNING, or CRITICAL. It also generates a summary of the readings.

6. Features

• View the current readings

• Add a valid whole-number reading from 0 to 100

• Classify readings into NORMAL, WARNING, or CRITICAL

• Display the total, average, minimum, and maximum readings

• Display the count and percentage for each category

• Show an overall status based on the readings

• Keep showing the menu until the user chooses Exit

7. Requirements

• Python 3

• A text editor or Python IDE

• A terminal or command prompt

8. How to Run

1. Save the program as `device_monitor.py`.

2. Open a terminal or command prompt in the folder containing the file.

3. Run:

```
python device_monitor.py
```

If your computer uses the `py` command, you can run `py device_monitor.py` instead.

4. Choose an option from the menu and follow the prompts.

9. Sample Output

This sample was captured by running the program and choosing option 4, then option 5.

```
MONITORING SUMMARY
Number of readings: 5
Total of readings: 160
Average temperature: 32.00 °C
Minimum temperature: 28 °C
Maximum temperature: 37 °C

NORMAL: 2 (40.00%)
WARNING: 2 (40.00%)
CRITICAL: 1 (20.00%)
Overall Status: ATTENTION REQUIRED
```

10. Test Cases

The test cases and observed console results are recorded in `test_cases.md`. Please run the cases on your own computer as a final check before submitting.

11. Known Limitations

• The readings are stored only while the program is running.

• The program does not save readings to a file or database.

• Only whole numbers from 0 to 100 are accepted as new readings.

12. Reflection

Before submitting, write this section in your own words. You can mention what you learned about functions, loops, input validation, lists, and calculating a summary. Also mention one part that was difficult and how you checked it.

3. test_cases.md

Test Cases — Simulated Device Monitoring and Alert System

Testing note: These observed results came from running the program with the listed console inputs. Please repeat the tests on your own computer and update the results if your output differs.

Test ID	Input / Steps	Expected Result	Actual Result Observed	Status
TC01	Choose `1`, then `5`	The five starting readings are displayed, then the program exits.	The starting readings include 28 °C through 29 °C (five readings total); the exit message is displayed.	Pass
TC02	Choose `2`, enter `33`; choose `1`, then `5`	The reading is accepted and added to the list.	“Reading added successfully.” appeared, and the list included the new 33 °C reading.	Pass
TC03	Choose `2`, enter `abc`; choose `1`, then `5`	Nonnumeric input is rejected and the readings stay unchanged.	“Invalid input. Enter a whole number only.” appeared; the list remained at the original five readings.	Pass
TC04	Choose `2`, enter `-5`; choose `1`, then `5`	The negative reading is rejected and the readings stay unchanged.	“Invalid reading. Enter a value from 0 to 100.” appeared; the list remained at the original five readings.	Pass
TC05	Choose `2`, enter `120`; choose `1`, then `5`	A value above 100 is rejected and the readings stay unchanged.	“Invalid reading. Enter a value from 0 to 100.” appeared; the list remained at the original five readings.	Pass
TC06	Choose `2`, enter `30`; choose `3`, then `5`	30 °C is classified as WARNING.	The program accepted 30 and displayed `30 °C -> WARNING`.	Pass
TC07	Choose `3`, then `5`	The reading 35 °C is classified as WARNING.	The program displayed `35 °C -> WARNING`.	Pass
TC08	Choose `4`, then `5`	The summary shows total 160, average 32.00, minimum 28, maximum 37, and ATTENTION REQUIRED.	Total 160, average 32.00 °C, minimum 28 °C, maximum 37 °C, and `ATTENTION REQUIRED` were displayed. Category counts were NORMAL 2, WARNING 2, and CRITICAL 1.	Pass
TC09	Enter `9`, then `5`	The program displays an invalid-choice message and shows the menu again.	“Invalid choice. Please enter a number from 1 to 5.” appeared, and the menu was shown again.	Pass
TC10	Choose `5`	The exit message is displayed and the program ends.	“Thank you for using the Device Monitoring System.” appeared and the program ended.	Pass

Bug Found / Hardest Test

Write this in your own words after reviewing or repeating the tests. Describe one issue you noticed and fixed, or explain which test was hardest to check and why. Do not claim a bug was fixed unless you actually found and fixed one.